

NATACHA SPITZER

Ph.D. Candidate in Economics, University of Pennsylvania

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Citizenship: France and United States

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Education

University of Pennsylvania	2021 – Present
<i>Ph.D. in Economics, expected May 2027 (Advisor: George Mailath)</i>	
Paris School of Economics	2019 – 2021
<i>Master of Analysis and Policy in Economics (Summa Cum Laude)</i>	
University of Rennes I	2016 – 2019
<i>Bachelor of Economics and Management (Summa Cum Laude)</i>	
<i>Exchange Program at the University of Rochester</i>	2018 – 2019

Research Interests

Field: Microeconomic Theory

Interests: Contract Theory, Repeated Games, Networks, Information Economics

Research

Contracting with Unknown Complementarity: Ratchets and Replacement Risk

Job Market Paper

Abstract: A principal and an agent contract over two periods under moral hazard. Neither knows the agent's complementarity with a new technology. The agent earns limited liability rents, and the principal can only commit to the current period. In each period, the cheapest implementation of strictly positive effort is a bonus contract, paying either on a single tail region of output or on both tails. A pure-strategy equilibrium exists and is unique for a sufficiently convex cost of effort. The agent's incentive to influence the principal's belief about the complementarity decomposes into an *extensive* retention force and an *intensive* ratchet force, measured against a manipulation-free benchmark. Where the two forces balance, pessimism in the monotone-likelihood-ratio order of the prior drives over-experimentation, while optimism leads to under-experimentation.

On the Irrelevance of Networks in Repeated Games

Working Paper, under review

Abstract: When do interaction and information networks affect equilibrium outcomes in environments with many bilateral repeated games? Using a multimarket Cournot counterexample with symmetric firms and markets, I show that the ability of all pairs of opponents to sustain collusion in equilibrium depends on both which players interact and what each player observes about others' interactions. I then develop a general framework and identify a sufficient condition on stage-game payoffs—*Optimality of Nash Reversion*—under which network structure is irrelevant: an action profile is sustainable between all opponents if and only if it is sustainable in the corresponding two-player game.

Optimal Monitoring of AI Use with Reputation Concerns

Working Paper, with Mengjia Xia

Abstract: A worker of privately known ability with reputation concerns chooses between costly effort and a free AI tool; a manager who cares about accuracy monitors his choice at a cost. The optimal policy is to monitor as little as possible or not at all: either everyone uses AI, or the manager monitors just enough that able workers exert effort and others use AI. Mandatory disclosure is never optimal. Optimal monitoring is non-monotonic in the quality of AI. Relative to the efficient benchmark, the outcome can involve too much or too little reliance on the tool. The policy has the same shape whether effort and AI are substitutes or complements. When monitoring can depend on the outcome, the optimal audit targets success when able workers are rare and failure when they are common.

Selection in Networks with Imperfect Observability

Master's Thesis, Paris School of Economics (Advisor: Francis Bloch; Reader: Olivier Compte)

Abstract: I study ranking and selection in a network in which agents observe cardinal information about their neighbors that a social planner does not. Under perfect observability, the planner can construct an ex-post incentive compatible and efficient ranking if and only if each agent has at least two neighbors, and the sparsest such networks are graphs with cyclic components. Under imperfect observability, efficiency is lost in Bayesian equilibrium in triangles, but an approximately efficient mechanism exists when signals are precise; welfare under the optimal mechanism increases in signal precision and is higher in directed than in undirected triangles.

Presentations

Econometric Society World Congress (Seoul, 2025); NSF Network Science in Economics Conference (Stanford, 2025, poster); Pennsylvania Economic Theory Conference (Penn State, 2024 and 2025, posters); Advanced School in Economic Theory (Israel Institute for Advanced Studies, 2023, poster); Penn Theory Lunch Seminar (2022–2026)

Other Research Experience

Research Assistant, Paris School of Economics	2019 – 2021
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to Francis Bloch, Nicolas Jacquemet, Hillel Rapoport, and Akiko Suwa-Eisenmann

Research Intern, CNRS-CREM, University of Rennes I	Summer 2019
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Supervisor: Marie-Hélène Hubert

Teaching Experience

Instructor	2023 – Present
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University of Pennsylvania

Game Theory	Summer 2025
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Intermediate Microeconomics	Summer 2023
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Teaching Assistant	2022 – Present
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University of Pennsylvania

Introduction to Microeconomics, Dr. Anne Duchene	Fall 2026
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Microeconomics for Managers, Prof. Kent Smetters	Summer 2026
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Behavioral Economics, Prof. Kevin He	Fall 2025
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Microeconomics for Managers, Prof. Kent Smetters	Summer 2025
Intermediate Microeconomics, Prof. George Mailath	Spring 2025
Industrial Organization, Prof. Aislinn Bohren	Fall 2024
Contract Theory, Prof. Steven Matthews	Spring 2024
Introduction to Microeconomics, Dr. Anne Duchene	Fall 2023
Mathematical Economics, Prof. Andrew Postlewaite	Spring 2023
Introduction to Microeconomics, Dr. Anne Duchene	Fall 2022

Head Tutor of Microeconomics

2017-2018

University of Rennes I

Awards

Sidney Weintraub Memorial Prize in Economics, University of Pennsylvania	2025
Robert and Anita Summer Graduate Research Award in Economics, University of Pennsylvania	2025
Fontaine Fellowship, University of Pennsylvania	2021

Additional Skills

Programming Languages: Python, R, Stata, MATLAB, Mathematica, LaTeX
Languages: English (Native), French (Native)

References

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Rakesh Vohra, Department of Economics, University of Pennsylvania
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